Joseph covariance retains a nonnegative excess

$\Sigma(B) = \Sigma(K) + (B - K)\, S\, (B - K)^{T}$. Coordinate final gain.

Chosen posterior $\Sigma(K)$

1.33333	-0.666667
-0.666667	0.833333

Joseph approximation $\Sigma(B_6)$

1.33336	-0.666661
-0.666661	0.833335

Exact covariance excess

2.26663e-05	5.66657e-06
5.66657e-06	1.41664e-06

Excess = [[16, 4], [4, 1]] / 705894

Exact identity independently checked via direct Joseph factors and rational inverse.